SOFTWARE REQUIREMENTS SPECIFICATION

AI-Based System-Wide Real-Time Masking of People (Linux desktop, paused)

A filter for your desktop that hides people on screen as they appear, in every app.

Paused, plan B (6 October 2026). The project moved to Android first. The current SRS is `srs-en.html`.

Version	Date	Scope
1.0 (draft)	3 October 2026	Linux v1

Status

For team review

1. Introduction 2. Overview 3. Features 4. Quality targets 5. Not in v1 6. Status & plan 7. Risks 8. Terms

1. Introduction

1.1 Purpose

This document explains what the filter must do and how well it must do it. It is written for the project team and supervisors, so it avoids deep technical detail. The full technical study and the test results live in `investigation-linux-v1.md` and `p0-results-linux-v1.md`.

1.2 The problem

Some people want to avoid seeing images of people while they use a computer: in social feeds, videos, ads, news and games. Today's tools record the screen, look at it, then draw a blur on top. That has three problems:

- The content is shown **before** it is checked, so it flashes on screen for a moment.
- They work like screen recording, which is heavy and asks for screen-sharing permission.

- They often work only in one app or one website.

1.3 Our idea

The filter sits inside the part of Linux that draws all windows on the screen. It sees each window's picture **before** it reaches the screen. Anything new stays hidden until it has been checked. Only then is it shown, with any people covered.

2. Overview

2.1 How it works, simply

- **The display part** runs inside the desktop. It takes a small copy of each window when it changes and applies the cover.
- **The detection part** is a separate background program. An AI model finds the people in that copy and sends back where they are.
- The two are separate on purpose. If detection crashes, the desktop keeps running and new content stays hidden.

Content changes	Covered while checked	Shown
new image, scroll, video frame	about 40–50 ms (2–3 frames)	with every detected person hidden

2.2 Users

One kind of user: a person who chooses to install the filter on their own computer. It is a personal tool, not a parental-control or monitoring tool.

2.3 Where it runs

ITEM	VERSION 1
System	Linux, with the Hyprland desktop (version 0.56)
Target laptop	Intel Core i5 (11th gen), built-in Intel Iris Xe graphics, 8 GB memory. A normal student laptop, no gaming graphics card.
Internet	Not needed. Everything runs on the device.

2.4 Assumptions

- The user runs Hyprland and agrees to install a plugin for it.
- "Person" means any human figure: real, drawn, animated or in a game.
- Hiding something by mistake is acceptable. Showing a person by mistake is not. When in doubt, hide.

3. Features (functional requirements)

Priority: **Must** = needed for v1, **Should** = planned for v1 if time allows, **Could** = later.

ID	REQUIREMENT	PRIORITY	STATUS
FR-1	Detect people in every open window: browser, video player, image viewer, games, chat apps, and the wallpaper.	Must	Tested offline
FR-2	Cover each detected person with a strong pixelation or solid fill, not a light blur. A light blur still shows the body shape.	Must	Tested offline
FR-3	Keep any new or changed content covered until it has been checked. Never show unchecked content.	Must	Planned
FR-4	Make the cover a little larger than the person so no edges show.	Must	Tested offline
FR-5	Do a closer second check on large windows (such as a full-screen feed) to find small or far-away people.	Must	Needs work
FR-6	Keep a person covered across frames even if the model misses them for a moment, so the cover does not flicker.	Must	Planned
FR-7	When the scene changes completely (a video cut, a new page), cover the whole window until it is checked.	Should	Planned
FR-8	If the detection program stops, cover all new content and show a small warning sign.	Must	Planned
FR-9	Apply the filter to screenshots and screen sharing too, so nothing leaks through them.	Must	Planned

FR-10	Let the user turn the filter on and off and choose how strong the cover is.	Should	Planned
FR-11	For video, delay the picture very slightly (up to 40 ms, too short to notice against the sound) instead of covering it.	Could	v1.1

4. Quality targets (non-functional requirements)

ID	AREA	TARGET	TODAY
NFR-1	Safety	Hide at least 95% of people in normal screen content. This is the most important number.	Not yet Large people: 86–95%. Small people in a busy feed: 35–69% with the closer check.
NFR-2	Speed	Changed content stays covered for no more than about 4 frames (about 65 ms).	Met About 40–50 ms in tests.
NFR-3	Smoothness	The desktop must not feel slower. Extra drawing work fits inside one screen frame.	Met in tests About 6 ms per frame. Not yet tested inside Hyprland.
NFR-4	Light on resources	No work when the screen is still. About 1 processor core and under 300 MB memory while content changes.	Met Battery use not measured yet.
NFR-5	Stability	A crash in detection must never freeze or close the desktop. Detection restarts by itself.	By design
NFR-6	Privacy	Screen images stay in memory only. Nothing is saved to disk and nothing is sent over the internet.	By design
NFR-7	Works on weak hardware	Still works, more slowly, on a laptop without usable graphics (processor only).	Met About 25 ms per check.
NFR-8	Future-ready	The AI model uses a standard format so it can move to Android later without retraining.	By design

5. Not in version 1

- Android and other phones
- Other Linux desktops (GNOME, KDE Plasma) and old X11 sessions
- Hiding only some people, for example by gender. This is much less accurate, and a mistake means a person is shown.
- Blocking the user from turning the filter off. It is a personal choice, not a lock.
- HDR screens and computers with more than one graphics card

6. Status and plan

STAGE	WHAT	TIME	STATUS
P0	Test AI models on the target laptop for speed and accuracy	days	Done
P1	Build the display part (Hyprland plugin) with fake detections	1–2 weeks	Next
P2	Build the detection program and connect the two parts	1–2 weeks	Planned
P3	Collect real screen recordings, label them with a large AI model, and train our model on them	weeks	Planned

What P0 showed: the laptop is fast enough. The open problem is accuracy on **small people**, such as thumbnails in a feed. Training on real screen content (P3) is the main fix.

7. Risks

RISK	EFFECT	WHAT WE DO
Small people are missed	Some people are shown	Closer second check, training on screen content, hide when unsure

Hyprland updates break the plugin	Filter stops working	Fix one Hyprland version. Keep the plugin very small.
A plugin bug	The whole desktop can crash	Put all heavy logic in the separate detection program
Weak laptop limits model size	Lower accuracy	Use a stronger model on stronger laptops

8. Terms

Frame	One picture on the screen. A normal screen shows 60 per second, so one frame is about 17 ms.
ms	Millisecond, one thousandth of a second.
Hyprland	The Linux desktop program that draws all windows on the screen. Our display part is a plugin for it.
AI model	The trained program that finds people in a picture.
Hide first	Our rule: new content stays covered until it is checked (in technical documents: "fail-closed").